LIGHTYEAR

Where context becomes trusted action.

MS 73 Run history bounded pattern reach and activity projection

Status: Implemented; engine recording remains MS74

Date: 2026-09-14

Audience: Operators, engineering teams and assurance reviewers

Executive summary

MS73 adds an index across finished runs and a read-only Control Tower view. The index retains one immutable row per run; customer-controlled compressed journals can be pruned under an explicit retention policy. The default is 396 days from completion. No pruning scheduler or journal storage service is installed.

Delivered capability

- Summarize completed, hash-chain-valid journals and bind each row to its journal path, byte count and SHA-256. This structural check is not semantic qualification; the engine must independently admit its runs before recording them in MS74.
- Rebuild all recorded row fields from surviving journals and their run/estate/ workload metadata. Retain index rows and hashes after pruning, with the actual pruning time. Pruning metadata cannot be reconstructed from deleted journals.
- Reject path traversal, changed journals, mismatched retention directories and replacement of existing run identities. Missing journals produce errors rather than successful deletion counts. Pruning is an operator-invoked function.
- Query weekly action and block events from index rows alone. The UI labels these as activity, not resolved findings or a measure of semantic convergence. Multiple actions or repeated runs can concern the same service.
- Reject invalid/empty patterns, empty registers and incompatible comparison records as unmeasured. A valid nonmatching register can produce a measured zero. Pattern reach does not execute normalization or establish an accepted difference.
- Serve `/api/workflow/convergence` before graph refresh. Reads never initialize or migrate a database. An absent index reports `no-runs-recorded`; an invalid index reports `invalid-run-index` and the UI reports unavailable history.

Evidence and acceptance

The original eleven supplied tests are retained, with the action count named accurately, complete rebuilt rows compared, journal reads prohibited during trend queries, and explicit SQLite connection cleanup. Additional regressions cover the real MS70 register, malformed counts, traversal, retention binding, immutable run identity, altered journals, read-only projections and HTTP dispatch ordering. Browser CI checks the empty and unavailable states on desktop and mobile and verifies that visiting the Tower never creates a run index.

The original `1252 passed, 21 skipped` count is superseded by these authorized regression additions. Actual suite counts and environment limitations are recorded in the PR; no test is removed to reproduce a historical total.

Boundaries and claims not made

- No customer run has produced a convergence trend. Test activity comes from the committed CloudBank journal; there is no seeded preview tool in the repository.
- Retention has only been exercised on disposable test journals. No customer journal has been pruned.
- `blast_radius` is not called by `planner.py`; emitted plans still report `suppression_estimate: not-assessed`. Raw MS70 pair-reference registers are rejected until a caller resolves their bound comparison evidence into the supported per-comparison register. Pattern matches alone do not authorize signing.
- The view has not been seen by a customer. Counts are action events, not unique resolved findings, current backlog, or proof of improving platform capability.
- No engine halt path calls `record()` yet. MS74 must supply verified completed runs and customer-controlled journal paths. The deployed view therefore remains empty after this change.
- The existing MS1–70 multi-format library remains intact. MS73 is a separately declared, content-addressed implementation record. Word and PDF companions were added in the October 14 documentation reconciliation.

Source of record

- Run index
- Pattern reach
- Regression tests
- Control Tower boundary